AI and Society

CSET-485

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UNIT - 1

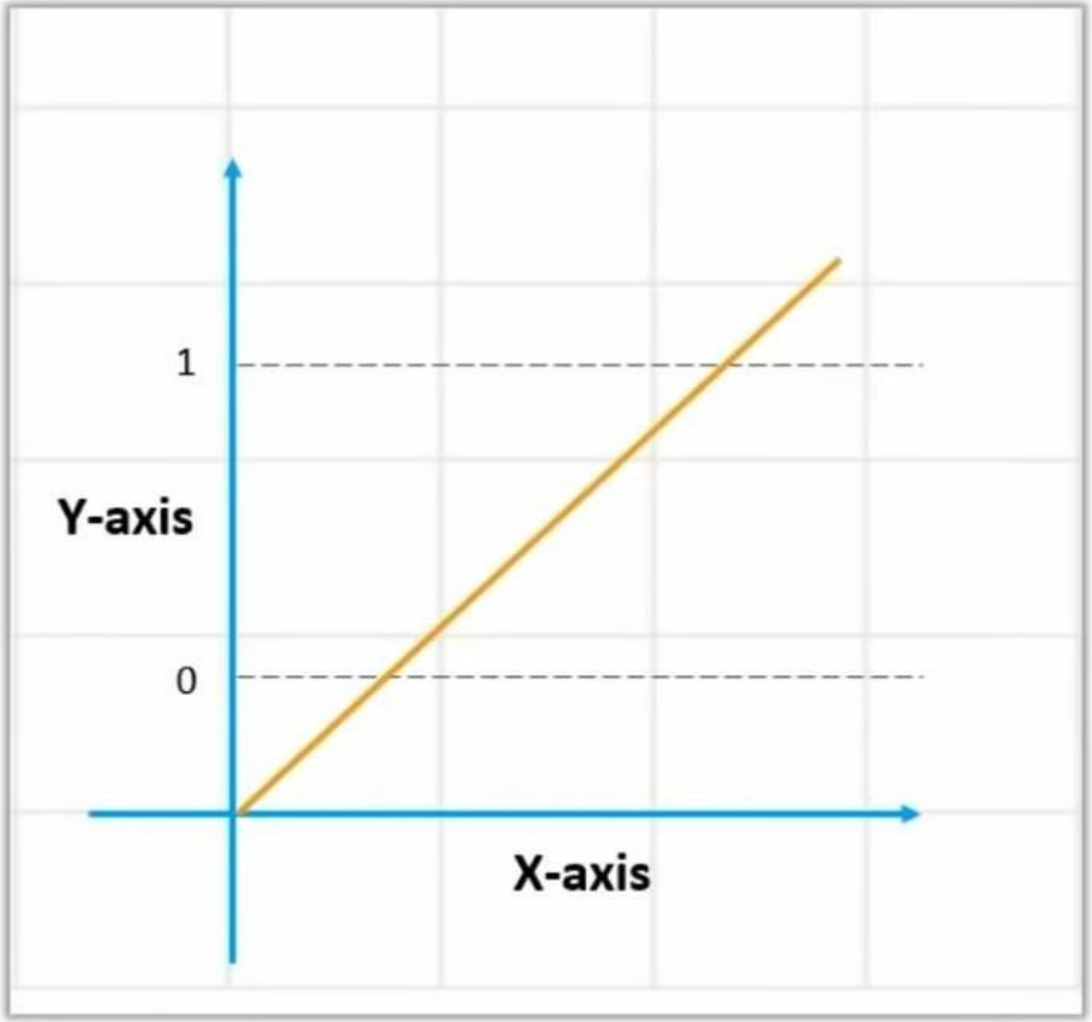
Logistic Regression

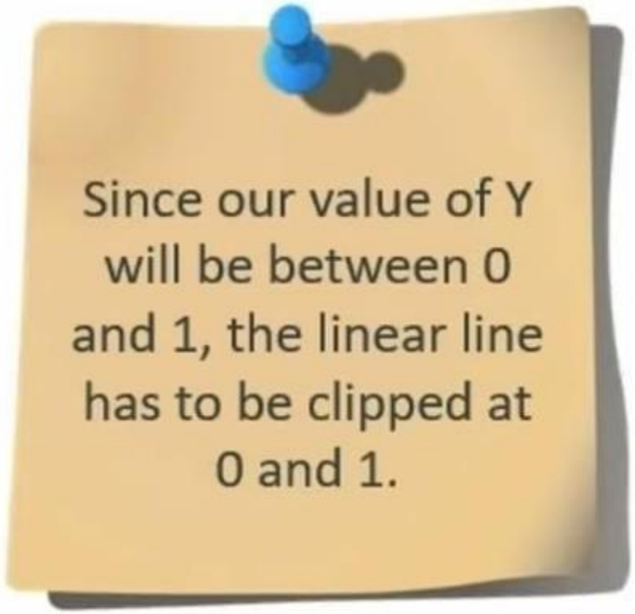
Logistic Regression

- Logistic Regression produces results in binary format which is used to predict the outcome of a categorical dependent variable.

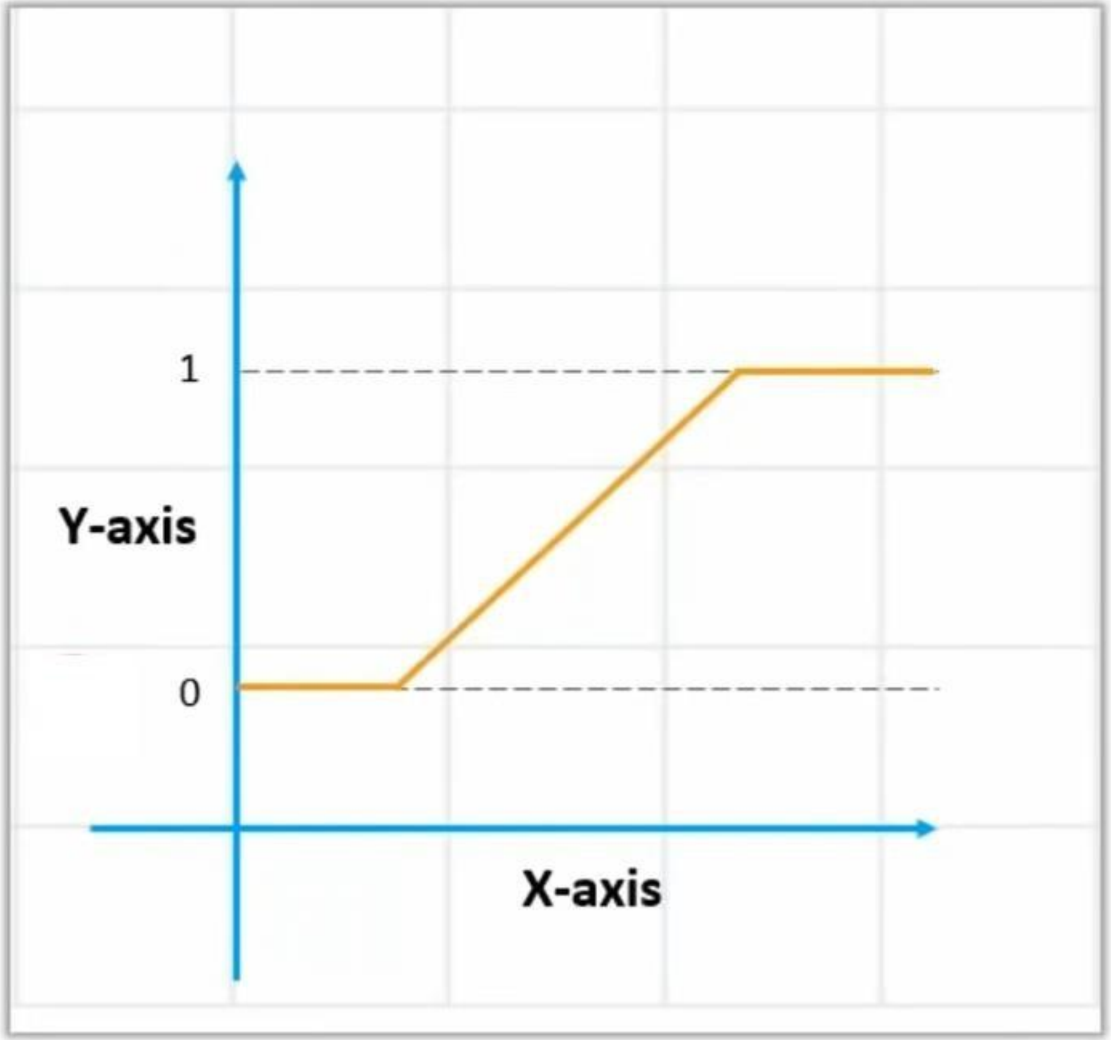
- So the outcome should be discrete/categorical.

Why not Linear Regression?

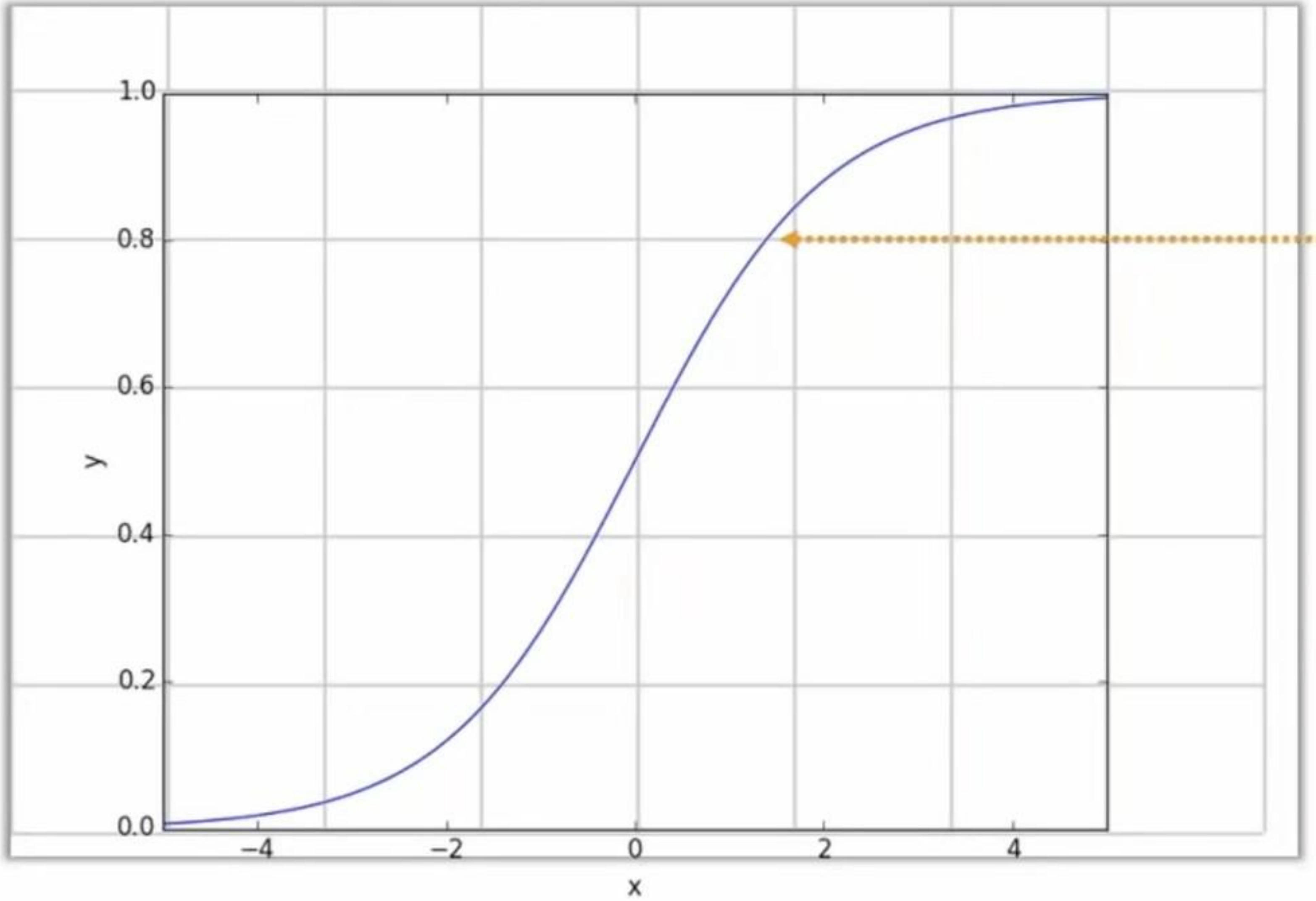




Why not Linear Regression?

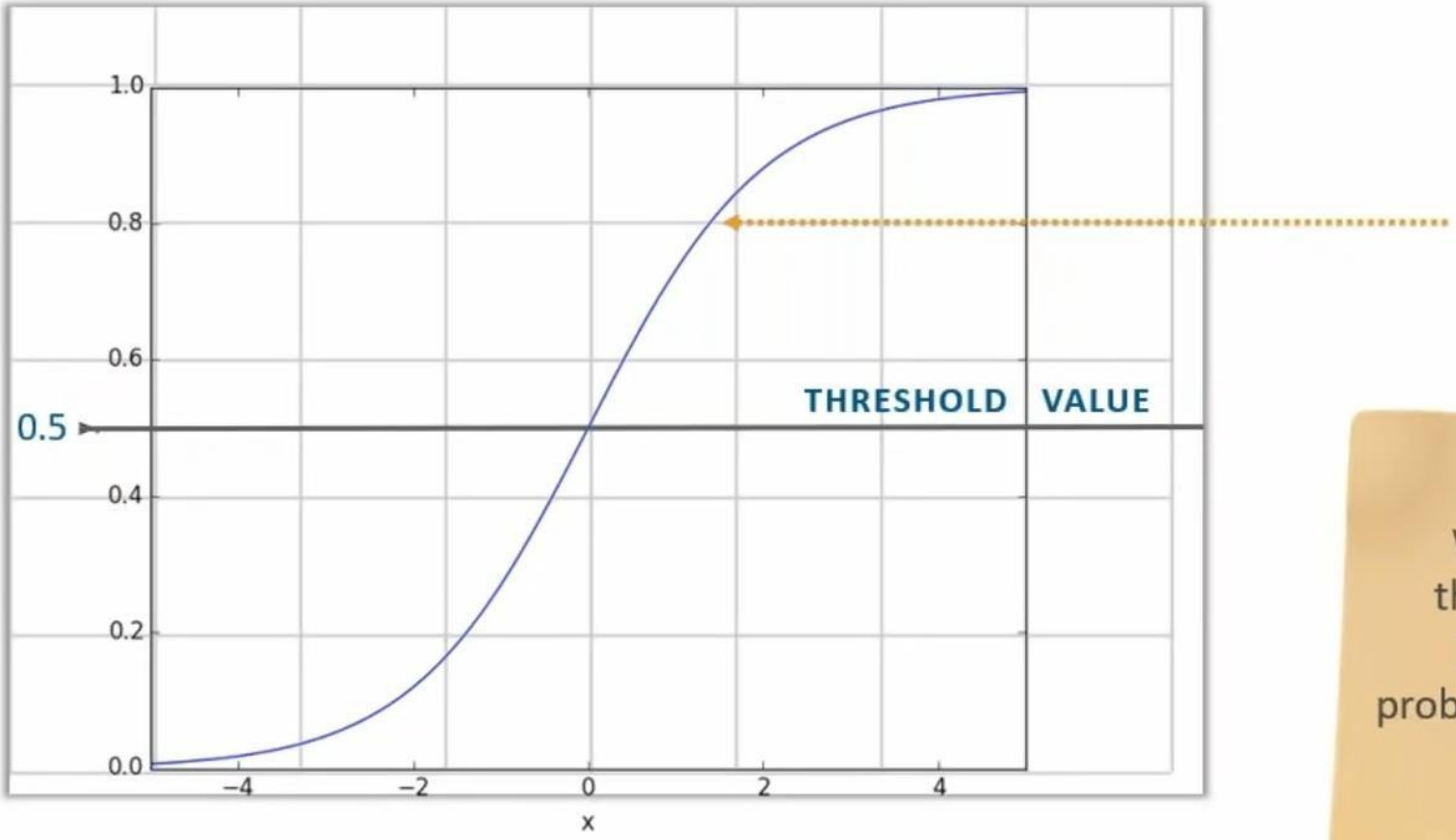


Logistic Regression Curve



The Sigmoid "S" Curve

Logistic Regression Curve



The Sigmoid "S" Curve

With this, the threshold value indicates the probability of winning or losing

Logistic Regression: Mathematical Details

- Logistic regression is a type of statistical model used to analyze the relationship between a binary outcome variable (dependent variable) and one or more predictor variables (independent variables).
- The binary outcome variable takes on one of two possible values, usually 0 or 1, indicating the absence or presence of an event, respectively.
- The logistic regression model assumes that the log-odds of the probability of the binary outcome variable, which is also called the logit, is a linear combination of the predictor variables.
- Mathematically, this can be expressed as:

$$\text{logit}(p) = \beta 0 + \beta 1x1 + \beta 2x2 + \ldots + \beta pxp$$

- where p is the probability of the binary outcome variable, β0 is the intercept, β1, β2, ..., βp are the coefficients for the predictor variables x1, x2, ..., xp, respectively.

Logistic Regression: Mathematical Details

- The logit function is defined as the natural logarithm of the odds of the event occurring, which is the probability of the event divided by the probability of its complement:

logit(p) = ln(p/(1-p))

- By solving for p in the above equation, we get the logistic function:

p = 1 / (1 + exp(-logit(p)))

- This logistic function has an S-shaped curve that ranges from 0 to 1, which represents the probability of the event occurring as a function of the predictor variables.

Logistic Regression: Derivation

- The probability that Y=1 given x is p, then the probability it does not happen is 1−p.
  Odds are defined as the ratio of the probability of success to failure

$$\text{odds} = \frac{p}{1-p}$$

- The odds range from 0 to +∞, which is not very convenient for modeling with linear functions (which range from −∞ to +∞)
  So, we take log of odds function, called logit function

$$\log\left(\frac{p}{1-p}\right)$$

- So, logistic regression assumes:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p \quad \text{expressed as} \quad \log\left(\frac{p}{1-p}\right) = \beta^\top \mathbf{x}$$

Logistic Regression: Derivation

- Logistic regression models the log-odds (logit function) as a linear function of the predictors

The left side is the natural logarithm (log base e) of the odds $\frac{p}{1-p}$.

So if $\log(A) = B$, then by definition of logarithms,

$$A = e^B$$

Exponentiate **both sides** to "undo" the log:

$$e^{\log\left(\frac{p}{1-p}\right)} = e^{\beta^\top \mathbf{x}}$$

Since $e^{\log(A)} = A$ (logarithm and exponent are inverse functions), this becomes:

$$\frac{p}{1-p} = e^{\beta^\top \mathbf{x}}$$

Multiply both sides by $(1-p)$:

$$p = (1-p)e^{\beta^\top \mathbf{x}}$$

$$p = e^{\beta^\top \mathbf{x}} - pe^{\beta^\top \mathbf{x}}$$

Bring terms involving p together:

$$p + pe^{\beta^\top \mathbf{x}} = e^{\beta^\top \mathbf{x}}$$

$$p(1 + e^{\beta^\top \mathbf{x}}) = e^{\beta^\top \mathbf{x}}$$

Linear Regression vs Logistic Regression

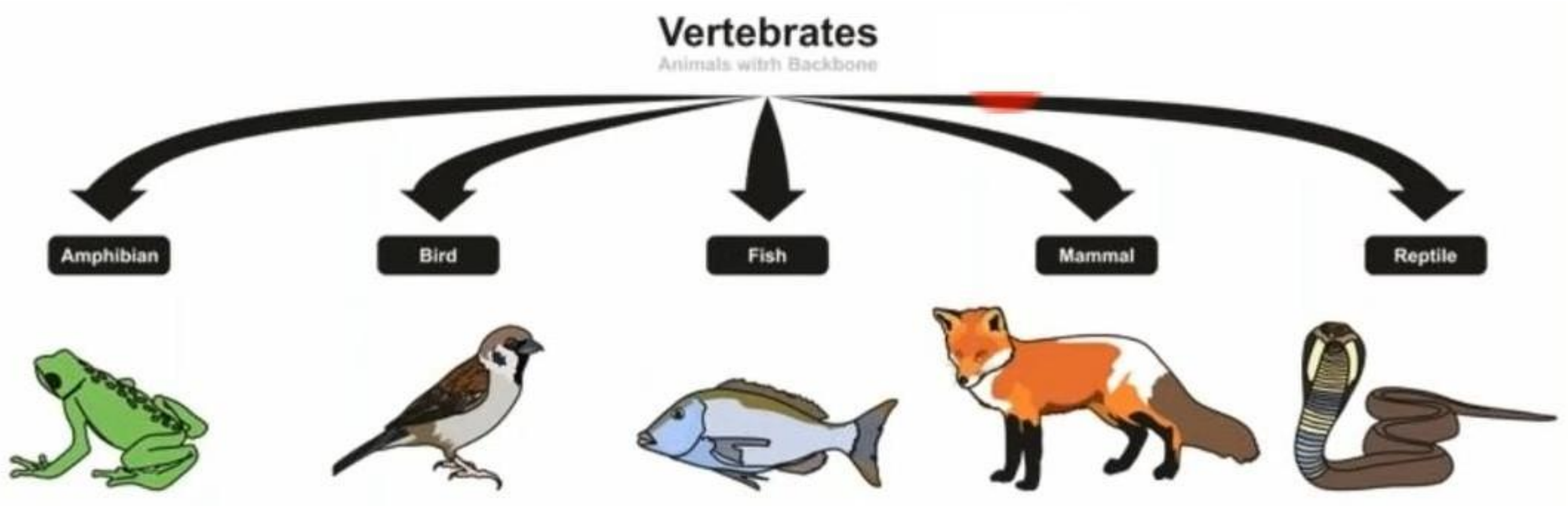
Linear Regression	Logistic Regression
1 Continuous variables	1 Categorical variables
2 Solves Regression Problems	2 Solves Classification Problems
3 Straight line	3 S-Curve

Applications of Logistic Regression: Weather predictions

- In weather predictions, logistic regression can be used to model the probability of certain weather events occurring, such as rainfall or snowfall, based on a set of predictor variables such as temperature, humidity, wind speed, and atmospheric pressure.

Applications of Logistic Regression: Classification Problems

- Logistic regression is commonly used in classification problems
- Logistic regression can be used to predict the probability of a particular species of vertebrate, such as a bird, fish, or mammal, based on a set of physical features such as beak shape, fin length, or body size.



Applications of Logistic Regression: Determine Illness

- Logistic regression can be used to predict the probability of a patient having a particular disease, based on a set of symptoms, medical history, or other clinical factors.

- For example, a logistic regression model could be built to predict the probability of a patient having diabetes, based on factors such as age, weight, family history, and blood sugar levels..

Evaluation Metrices: Confusion Matrix

- A confusion matrix is a table that is often used to describe the performance of a classification model (or "classifier") on a set of test data for which the true values are known.

- Consider binary classification:

n=165	**Predicted: NO**	**Predicted: YES**
Actual: NO	50	10
Actual: YES	5	100

Evaluation Metrices: Confusion Matrix

- There are two possible predicted classes: "yes" and "no".
- If we were predicting the presence of a disease, for example, "yes" would mean they have the disease, and "no" would mean they don't have the disease.
- The classifier made a total of 165 predictions (e.g., 165 patients were being tested for the presence of that disease).
- Out of those 165 cases, the classifier predicted "yes" 110 times, and "no" 55 times.
- In reality, 105 patients in the sample have the disease, and 60 patients do not.

n=165	Predicted: NO	Predicted: YES
Actual: NO	50	10
Actual: YES	5	100

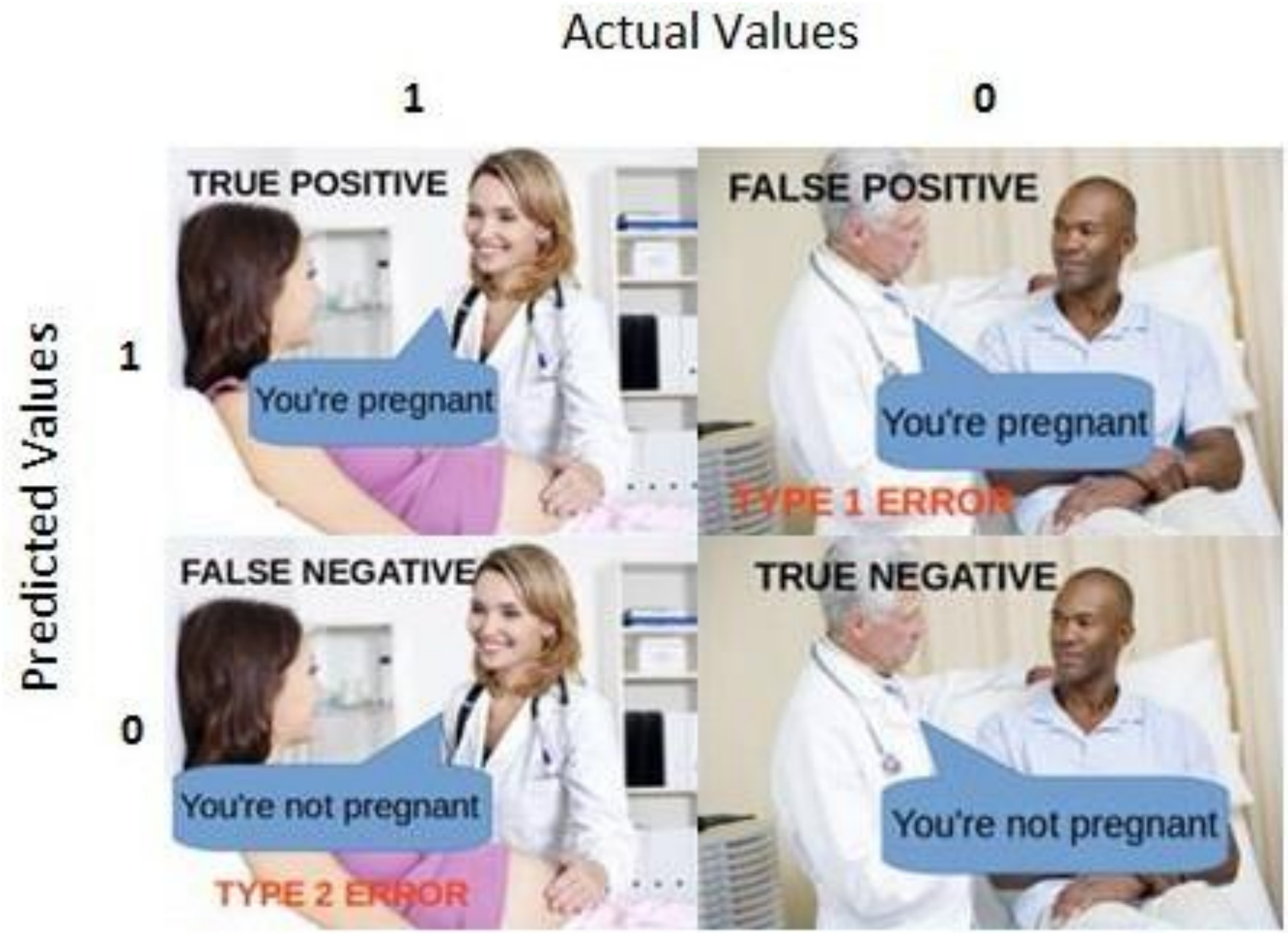
Evaluation Metrices: Confusion Matrix

- **True positives (TP):** These are cases in which we predicted yes (they have the disease), and they do have the disease.
- **True negatives (TN):** We predicted no, and they don't have the disease.
- **False positives (FP):** We predicted yes, but they don't actually have the disease **("Type I error")**
- **False negatives (FN):** We predicted no, but they actually do have the disease **("Type II error")**

n=165	**Predicted: NO**	**Predicted: YES**
Actual: NO	50	10
Actual: YES	5	100

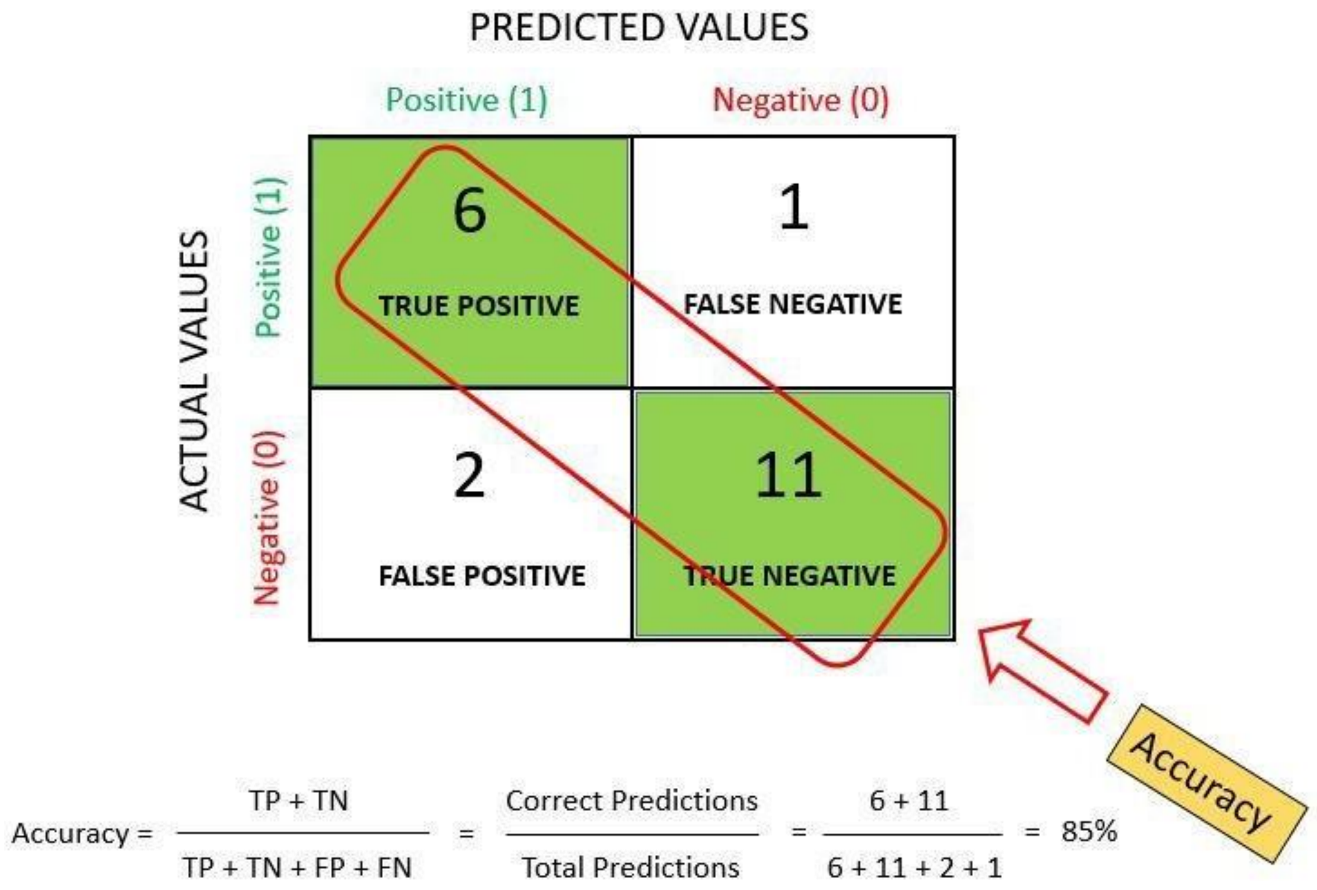
n=165	**Predicted: NO**	**Predicted: YES**	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

Evaluation Metrices: Confusion Matrix



Evaluation Metrices: Accuracy

- Accuracy simply measures how often the classifier makes the correct prediction. It’s the ratio between the number of correct predictions and the total number of predictions.



Evaluation Metrices: Precision

- It is a measure of **correctness** that is achieved in **true prediction**. In simple words, it tells us how many predictions are ***actually positive*** out of all the ***total positive predicted***.

- Precision is defined as the ratio of the total number of *correctly classified positive classes* divided by the *total number of predicted positive classes*.

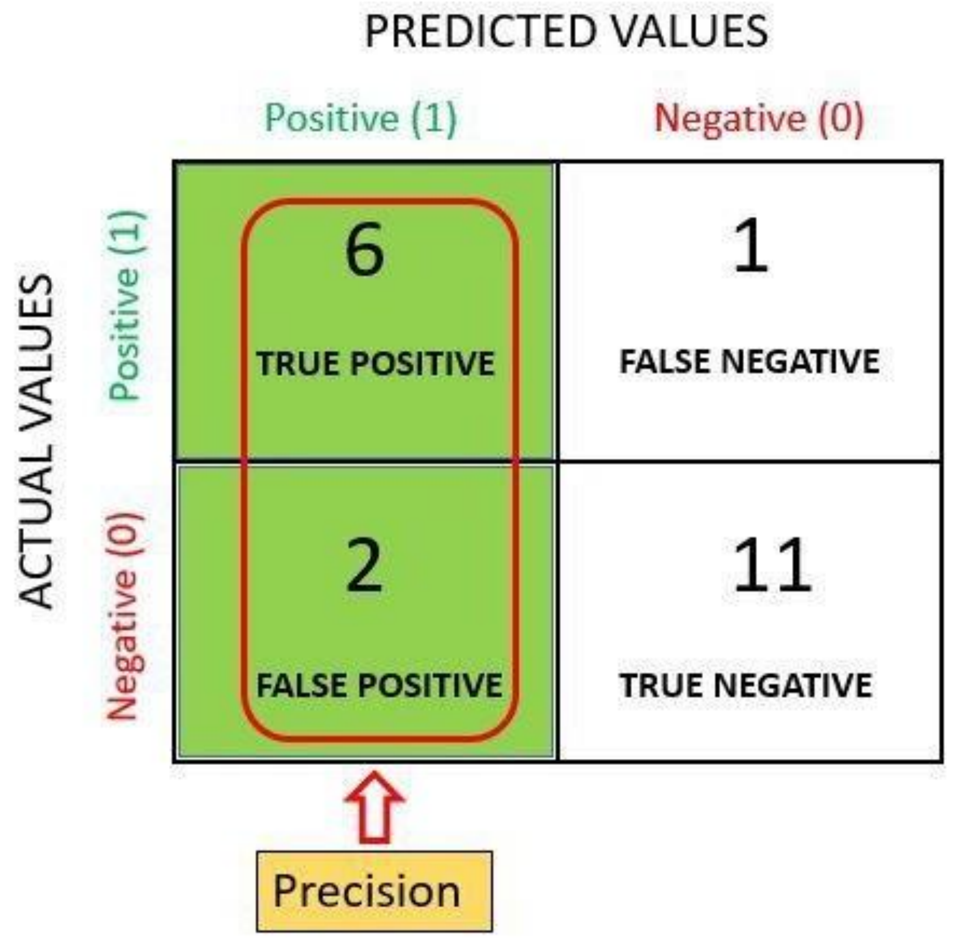
- Or, out of all the predictive positive classes, how much we predicted correctly. **Precision should be high(ideally 1).**

- *"**Precision** is a useful metric in cases where **False Positive** is a higher concern than **False Negatives**"*

- **Ex 1:- In Spam Detection** : Need to focus on **precision**

- Suppose **mail is not a spam** but model is **predicted as spam :** FP (False Positive). We always try to reduce FP.

Evaluation Metrices: Precision



$$\text{Precision} = \frac{TP}{TP + FP} = \frac{\text{Predictions Actually Positive}}{\text{Total Predicted positive}} = \frac{6}{6+2} = 0.75$$

Evaluation Metrices: Recall

- “It is a measure of **actual observations** which are predicted **correctly**, i.e. how many observations of positive class are actually predicted as positive.

- ***Recall*** is a valid choice of evaluation metric when we want to capture ***as many positives*** as possible.

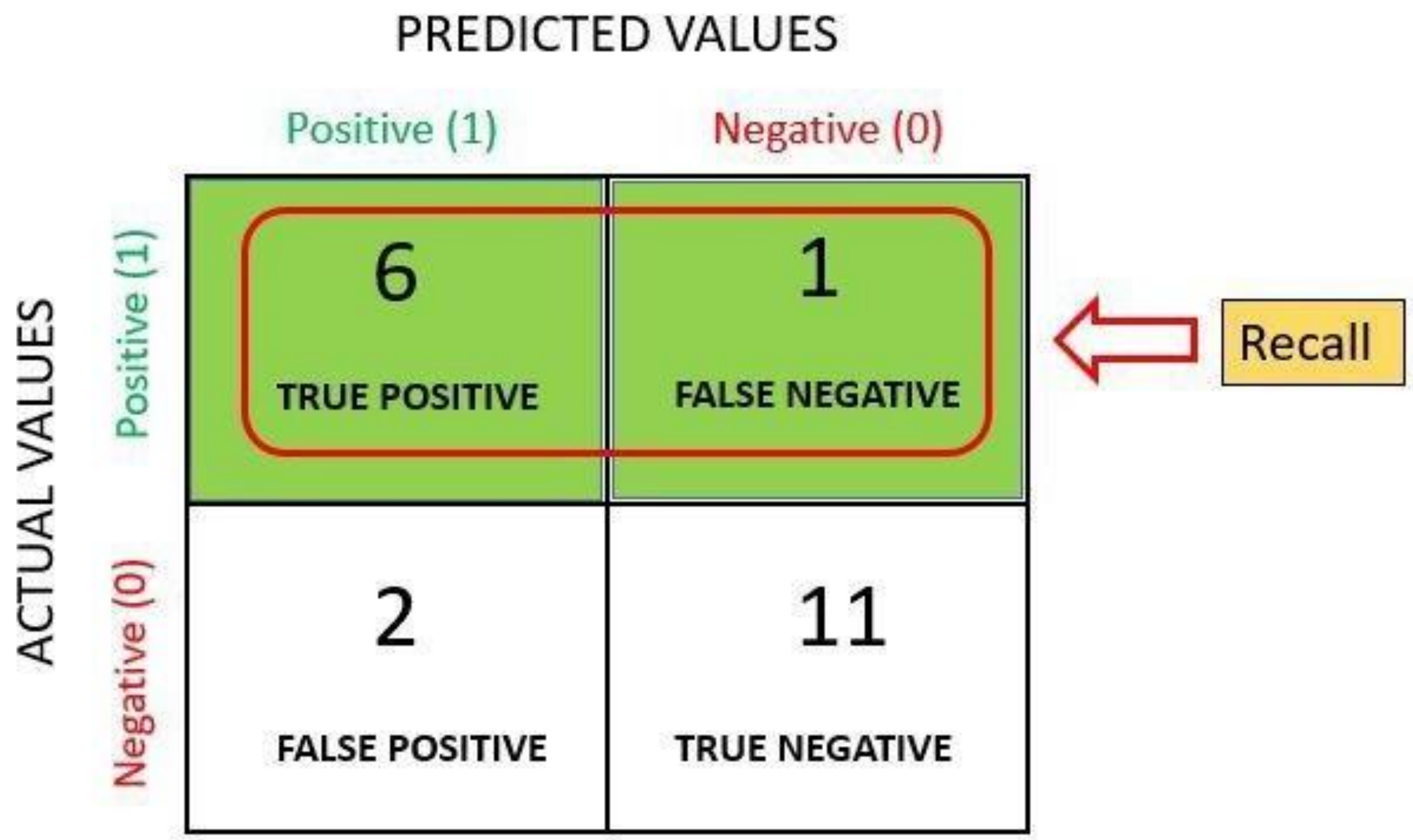
- Recall is defined as the ratio of the total number of *correctly classified positive classes* divide by the *total number of positive classes*. Or, out of all the positive classes, how much we have predicted correctly. **Recall should be high(ideally 1).**

- *“**Recall** is a useful metric in cases where **False Negative** trumps **False Positive**”*

- ***Ex 1:-*** suppose person having cancer (or) not? **He is suffering from cancer** but model predicted as **not suffering from cancer**

- *Recall would be a better metric because we don’t want to **accidentally discharge an infected person** and let them mix with the healthy population thereby **spreading contagious virus**. Now you can understand why accuracy was a bad metric for our model.*

Evaluation Metrices: Recall



$$\text{Recall} = \frac{TP}{TP + FN} = \frac{\text{Predictions Actually Positive}}{\text{Total Actual positive}} = \frac{6}{6 + 1} = 0.85$$

Evaluation Metrices: F-measure / F1-Score

- The **F1 score** is a number between **0 and 1** and is the ***harmonic mean of precision and recall***.

- **F1 score** sort of maintains a **balance** between the ***precision and recall*** for your classifier. If your ***precision is low***, the ***F1 is low*** and if the ***recall is low*** again your ***F1 score is low***.

- There will be cases where there is no clear distinction between whether *Precision is more important or Recall*. We combine them!

$$\text{F1-Score} = 2 * \frac{(\text{Recall} * \text{Precision})}{(\text{Recall} + \text{Precision})} = 2 * \frac{(0.85 * 0.75)}{(0.85 + 0.75)} = 0.79$$

Why so many evaluation metrices?

- Is it necessary to check for recall (or) precision if you already have a high accuracy?

Why so many evaluation metrices?

- Is it necessary to check for recall (or) precision if you already have a high accuracy?
- **We can not rely on a single value of accuracy in classification when the classes are imbalanced.**
- For example, we have a dataset of 100 patients in which 5 have diabetes and 95 are healthy. However, if our model only predicts the majority class i.e. all 100 people are healthy even though we have a classification accuracy of 95%.

When to use Accuracy / Precision / Recall / F1-Score?

- ***Accuracy*** is used when the ***True Positives and True Negative***s are more important. ***Accuracy*** is a better metric for ***Balanced Data.***
- Whenever **False Positive** is much more important use ***Precision.***
- Whenever **False Negative** is much more important use ***Recall.***
- ***F1-Score*** is used when the ***False Negatives and False Positives*** are important. ***F1-Score*** is a better metric for ***Imbalanced Data.***